

(12) **United States Patent**
Kwon et al.

(10) **Patent No.:** **US 12,399,429 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **SEMICONDUCTOR FABRICATION APPARATUS**

USPC 430/4, 5
See application file for complete search history.

(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(56)

References Cited

(72) Inventors: **Ohkug Kwon**, Yongin-si (KR); **Sanghun Kim**, Seoul (KR); **Junghwan Kim**, Bucheon-si (KR); **Sangbeom Park**, Hwaseong-si (KR); **Youngduk Suh**, Seoul (KR); **Jinwook Jung**, Suwon-si (KR); **Kukbin Choi**, Hwaseong-si (KR); **Dongkyeng Han**, Suwon-si (KR); **Tae-Hyun Han**, Hwaseong-si (KR)

U.S. PATENT DOCUMENTS

4,443,657 A	4/1984	Hill et al.
5,245,134 A	9/1993	Vana, Jr. et al.
5,502,287 A	3/1996	Nguyen
7,670,639 B2	3/2010	Gallant et al.
8,950,714 B2	2/2015	Ikeda et al.
10,518,511 B2	12/2019	Malloy et al.
2005/0024621 A1*	2/2005	Korenaga G03F 7/70716 355/75

(Continued)

(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 495 days.

JP	6090099	3/2017
KR	10-1970827	4/2019
WO	90006010	5/1990

Primary Examiner — Caleen O Sullivan

(74) *Attorney, Agent, or Firm* — Morgan, Lewis & Bockius LLP

(21) Appl. No.: **17/935,157**

(22) Filed: **Sep. 26, 2022**

(65) **Prior Publication Data**

US 2023/0251576 A1 Aug. 10, 2023

(30) **Foreign Application Priority Data**

Feb. 8, 2022 (KR) 10-2022-0015922

(51) **Int. Cl.**
G03F 7/20 (2006.01)

(52) **U.S. Cl.**
CPC **G03F 7/2002** (2013.01); **G03F 7/2045** (2013.01)

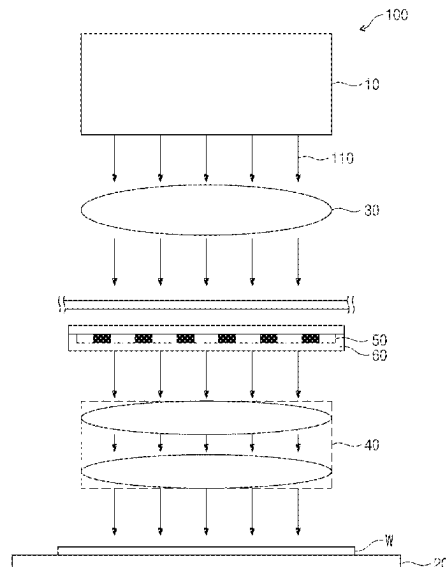
(58) **Field of Classification Search**
CPC .. G03F 7/2002; G03F 7/2045; G03F 7/70758; G03F 7/70991; G03F 7/70716

(57)

ABSTRACT

A semiconductor fabrication apparatus comprising a light source configured to emit light, a substrate stage arranged to receive a substrate exposed to the emitted light, a reticle arranged between the substrate stage and the light source, and a reticle stage arranged to receive the reticle. The reticle stage including a lower plate, an upper plate arranged above the lower plate, an actuator connected to the lower plate configured to move in a direction parallel to the upper plate, a first cable slab arranged between the upper plate and the lower plate and connected to one side of the actuator, and a first cable cover that surrounds an outer periphery of the first cable slab and contacts the lower plate when the first cable slab becomes bent.

20 Claims, 7 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0187439	A1 *	8/2006	Korenaga	G03F 7/70716 355/75
2007/0002785	A1 *	1/2007	Heo	G03F 7/70716 370/321
2017/0108785	A1 *	4/2017	Aoki	G03F 7/70816
2020/0012203	A1 *	1/2020	Cong	G03F 7/70758
2020/0117103	A1 *	4/2020	Cong	G03F 7/20

* cited by examiner

FIG. 1

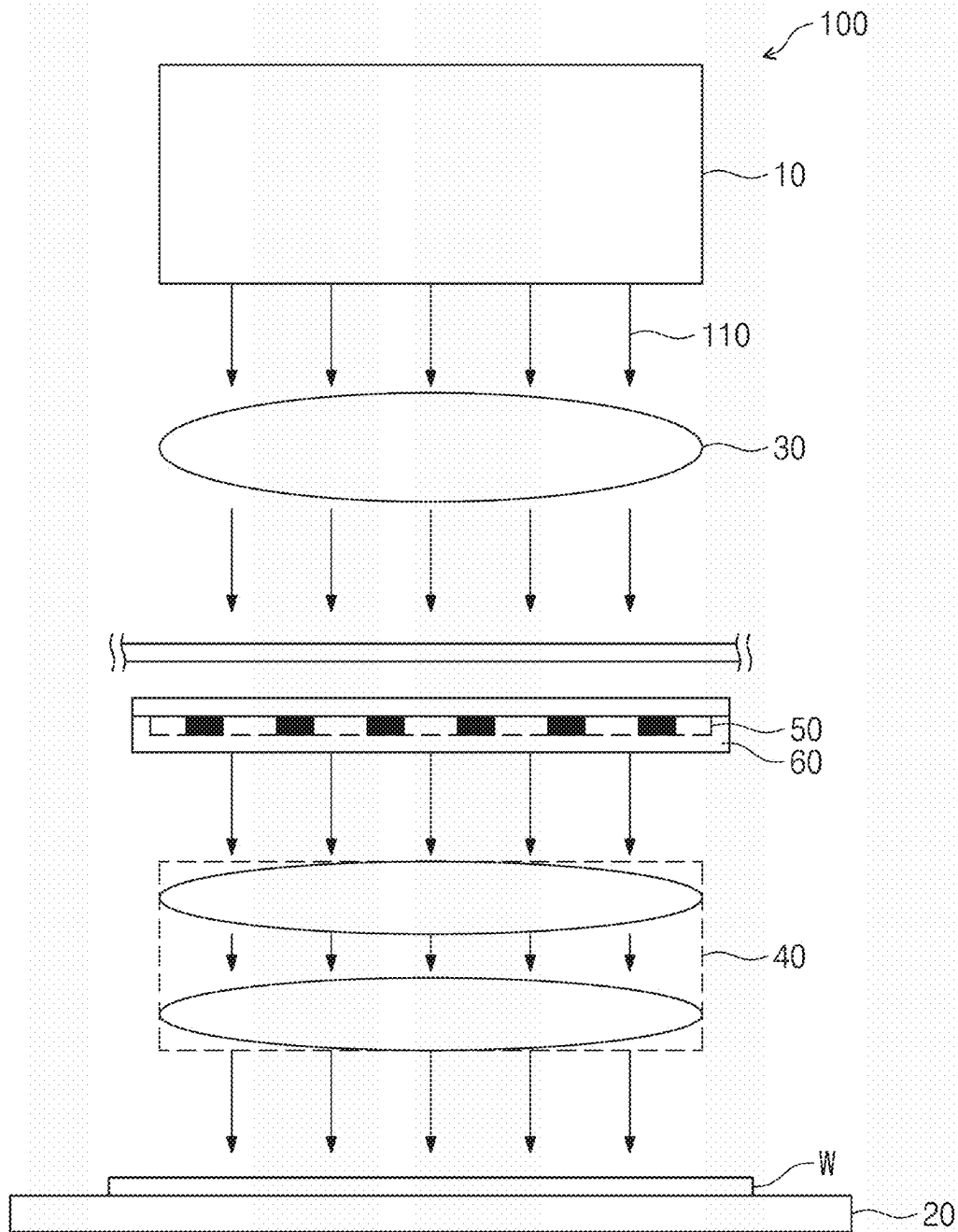


FIG. 2

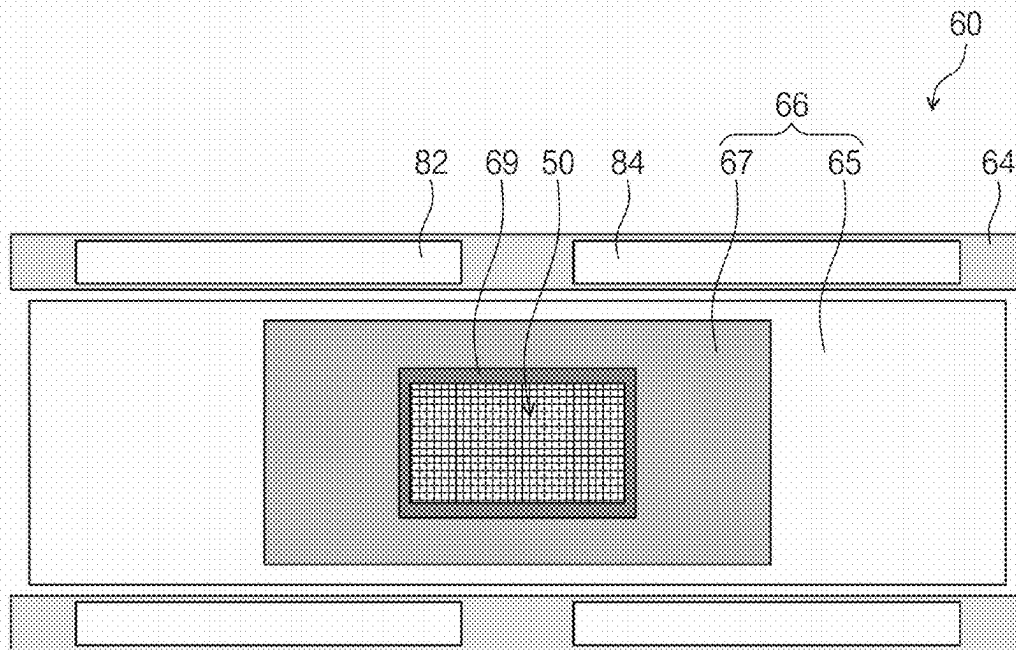


FIG. 3

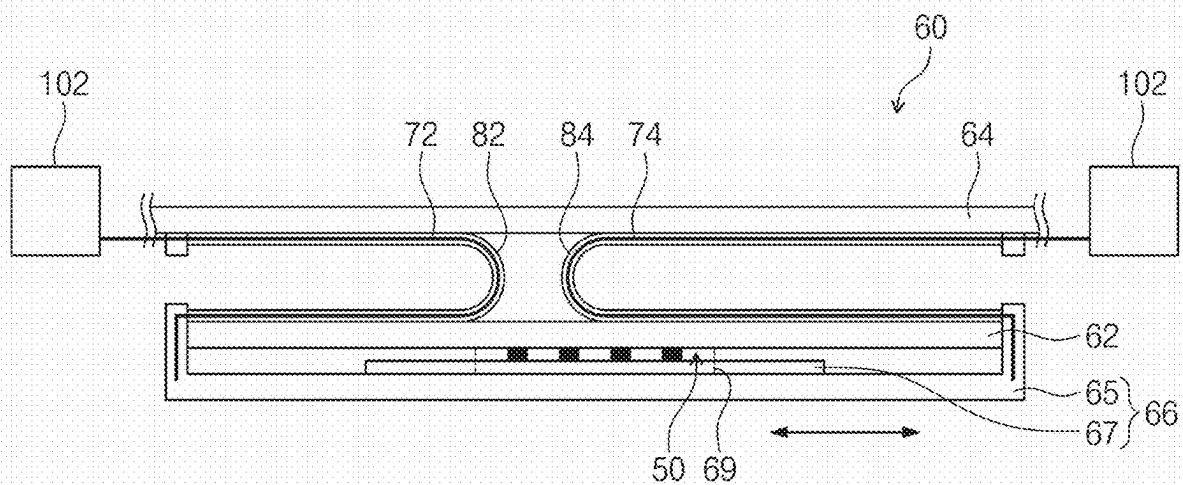


FIG. 4

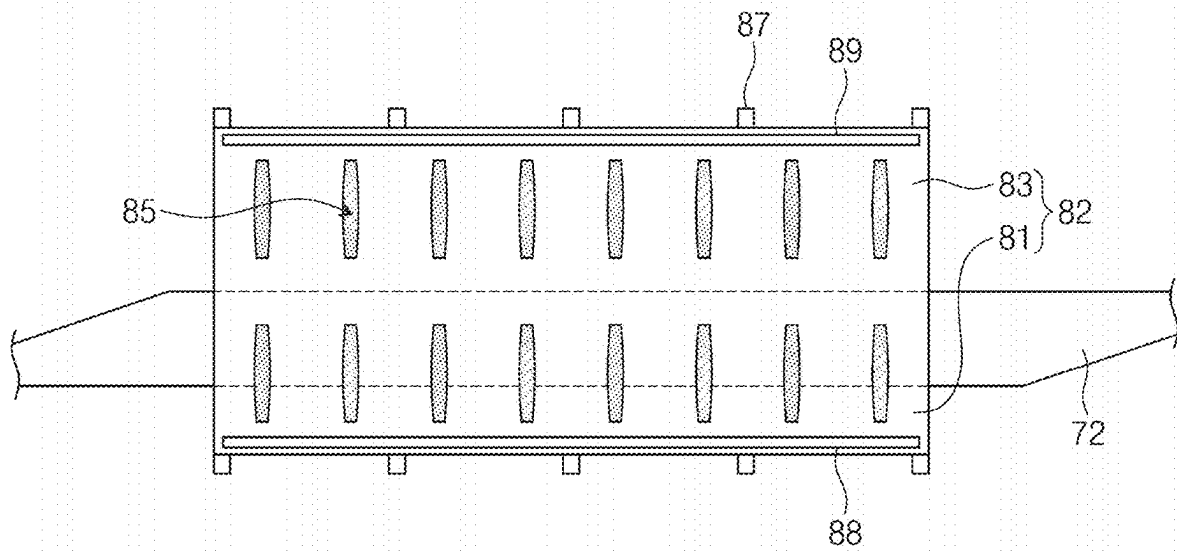


FIG. 5

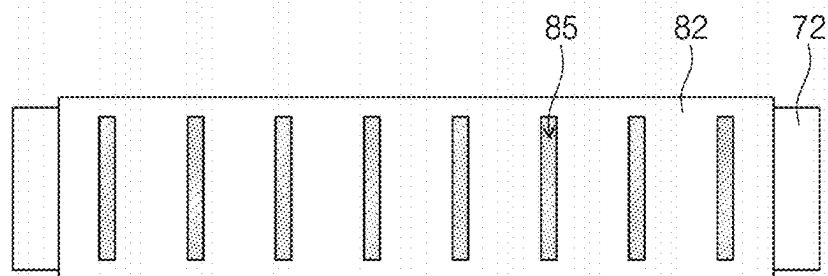


FIG. 6

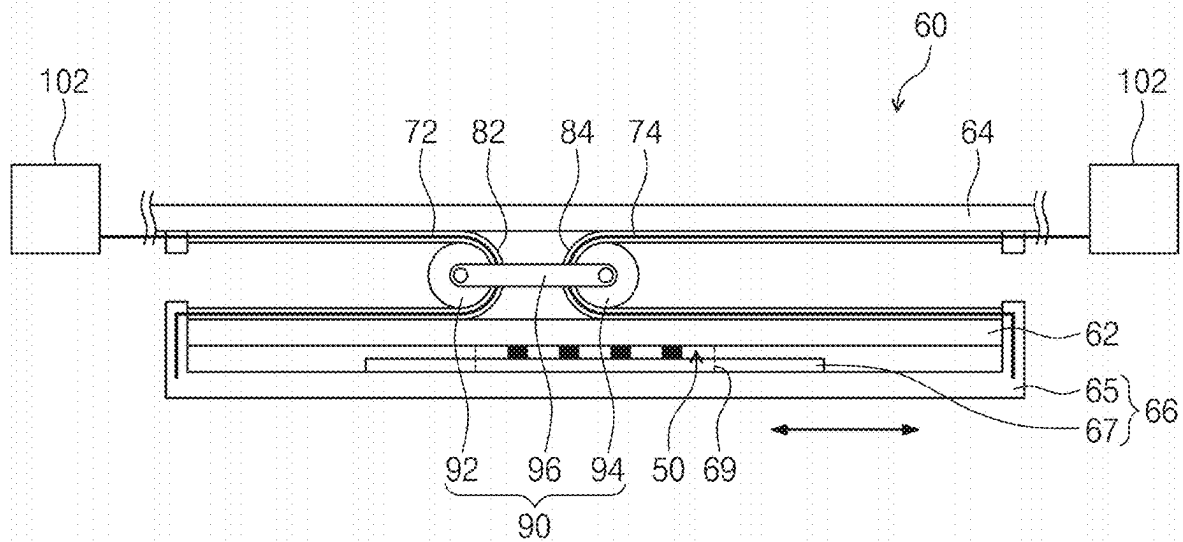


FIG. 7

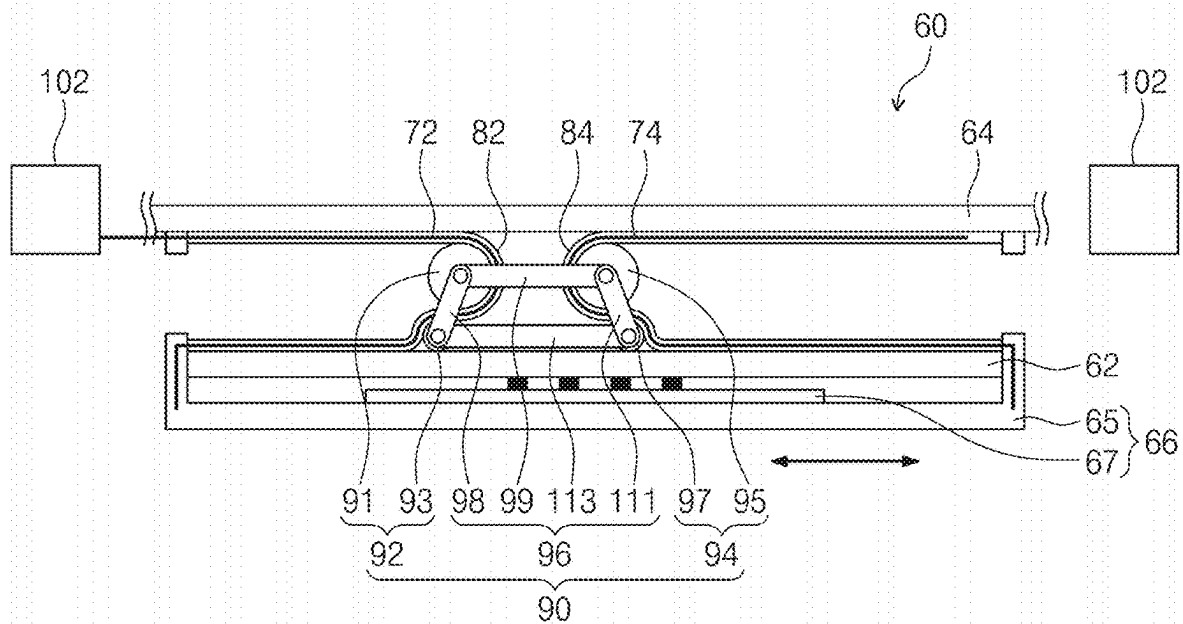


FIG. 8

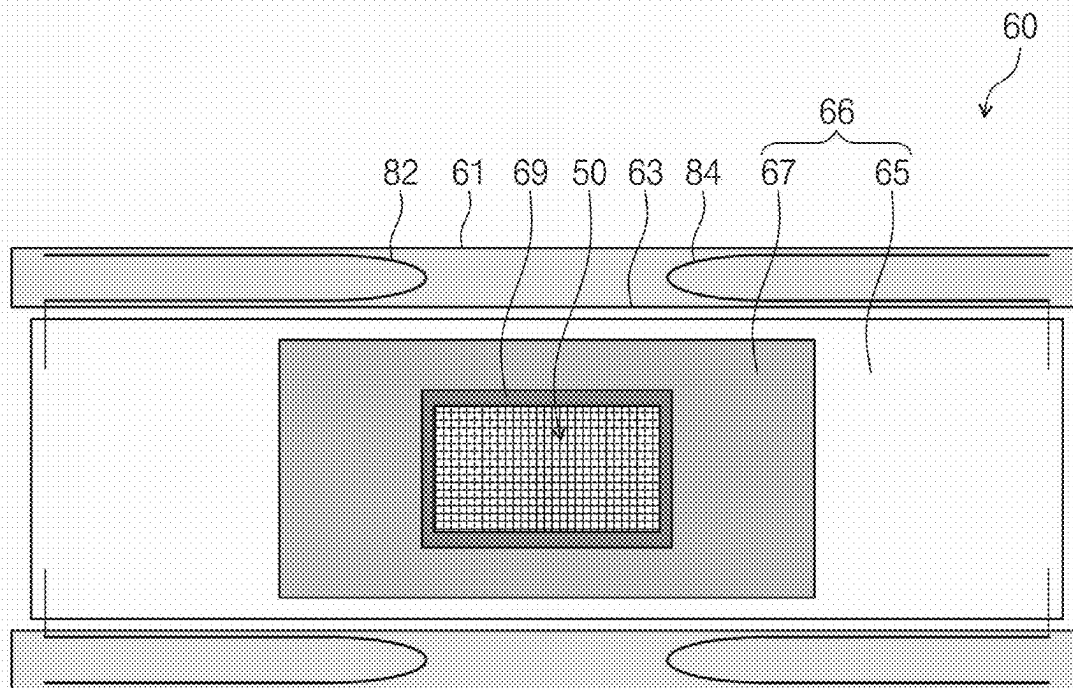


FIG. 9

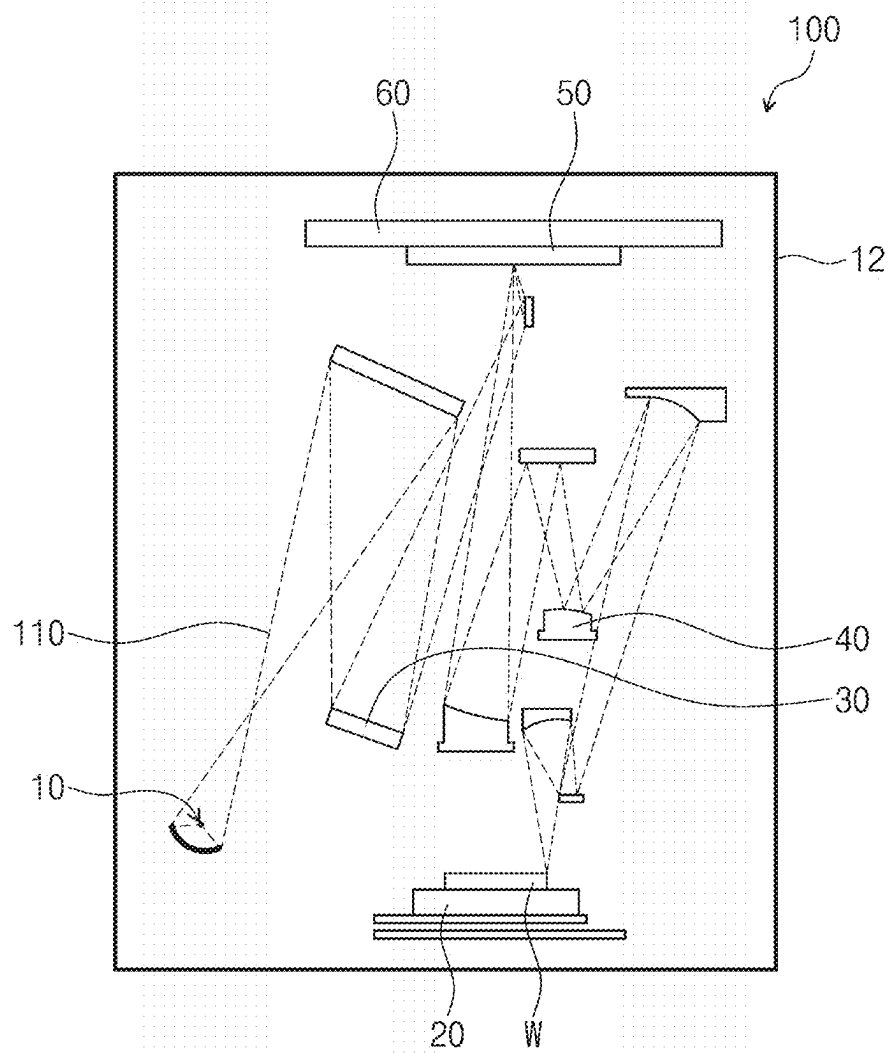
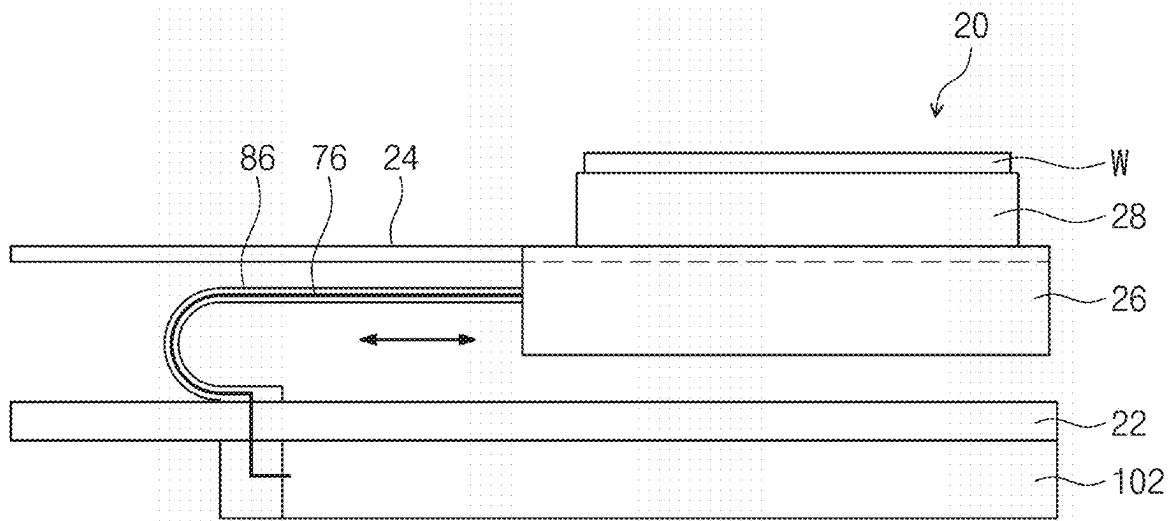


FIG. 10



1

SEMICONDUCTOR FABRICATION APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2022-0015922 filed on Feb. 8, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments of the present inventive concept relate to a semiconductor fabrication apparatus, and more particularly, to a semiconductor fabrication apparatus used for photolithography.

DISCUSSION OF RELATED ART

In general, a photolithography process may include a coating process, an exposure process, and a developing process. The exposure process may be a process that transports an image pattern onto a photoresist on a substrate. With an increase in the integration of semiconductor devices, research on the exposure process is being conducted to reduce a wavelength of light used in the exposure process to transport an image pattern onto a photoresist on a substrate. A typical semiconductor fabrication apparatus may include a reticle stage and a substrate stage. Each of the reticle stage and the substrate stage may have a cable slab connected to a controller.

SUMMARY

Embodiments of the present inventive concept provide a semiconductor fabrication apparatus capable of preventing or minimizing abrasion of cable slab.

According to some embodiments of the present inventive concept, a substrate fabrication apparatus may comprise: a light source configured to emit light; a substrate stage that receives a substrate exposed to the light; a reticle arranged between the substrate stage and the light source; and a reticle stage that receives the reticle. The reticle stage may include: a lower plate; an upper plate above the lower plate; an actuator connected to the lower plate, arranged to move in a direction parallel to the upper plate; a first cable slab arranged between the upper plate and the lower plate, the first cable slab being connected to one side of the actuator; and a first cable cover that surrounds an outer periphery of the first cable slab and contacts the lower plate when the first cable slab is bent.

According to some embodiments of the present inventive concept, a semiconductor fabrication apparatus may comprise: a chamber; a light source in the chamber configured to emit light; a substrate stage arranged to receive a substrate exposed to the light; a reticle arranged between the substrate stage and the light source; and a reticle stage that receives the reticle. The substrate stage may include: a lower plate; an upper plate on the lower plate; an actuator connected to the lower plate, actuator moving in a direction parallel to the upper plate; a cable slab arranged between the upper plate and the lower plate, the cable slab being connected to one side of the actuator; and a cable cover that surrounds an outer periphery of the cable slab, the cable cover contacting the lower plate when the cable slab is bent.

2

According to some embodiments of the present inventive concept, a semiconductor fabrication apparatus may comprise: a light source configured to emit light; a substrate stage that receives a substrate exposed to the light; a reticle between the substrate stage and the light source; and a reticle stage that receives the reticle. The reticle stage may include: a first lower plate; a first upper plate on the first lower plate; a first actuator connected to the first lower plate and moving in a direction parallel to the first upper plate; a first cable slab between the first upper plate and the first lower plate and connected to one side of the first actuator; a first cable cover that surrounds an outer periphery of the first cable slab and contacts the first lower plate when the first cable slab is bent; a second cable slab between the first upper plate and the first lower plate and connected to another side of the first actuator; a second cable cover that surrounds an outer periphery of the second cable slab and contacts the first lower plate when the second cable slab is bent; and a roller vehicle on the first cable slab and the second cable slab, the roller vehicle moving together with the first actuator.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 illustrates a cross-sectional view of a semiconductor fabrication apparatus according to an embodiment of the present inventive concept.

FIGS. 2 and 3 illustrate a plan view and a cross-sectional view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

FIGS. 4 and 5 respectively illustrate a development view and a plan view of a first cable cover of the reticle stage as depicted in FIGS. 2 and 3.

FIG. 6 illustrates a cross-sectional view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

FIG. 7 illustrates a cross-sectional view of a roller vehicle of the reticle stage as depicted in FIG. 6.

FIG. 8 illustrates a plan view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

FIG. 9 illustrates a cross-sectional view of a semiconductor fabrication apparatus according to an embodiment of the present inventive concept.

FIG. 10 illustrates a cross-sectional view of a substrate stage of the semiconductor fabrication apparatus as depicted in FIG. 9.

DETAIL DESCRIPTION OF EMBODIMENTS

Exemplary embodiments of the present inventive concept will be described more fully hereinafter with reference to the accompanying drawings. Like reference numerals may refer to like elements throughout the accompanying drawings.

FIG. 1 shows an example of a semiconductor fabrication apparatus 100 according to an embodiment of the present inventive concept.

Referring to FIG. 1, a semiconductor fabrication apparatus 100 of the present inventive concept may be embodied, for example, as a scanner exposure device. Alternatively, the semiconductor fabrication apparatus 100 may be embodied as a stepper exposure device. For example, the semiconductor fabrication apparatus 100 may be an ArF exposure device or a KrF exposure device. According to an embodiment, the semiconductor fabrication apparatus 100 may include a light

source **10**, a substrate stage **20**, an illumination optical system **30**, a projection optical system **40**, a reticle **50**, and a reticle stage **60**.

The light source **10** may produce and provide light **110** to be emitted towards a substrate **W**. In an embodiment, the light **110** may include excimer laser light or deep ultraviolet light. For example, the light **110** may have a wavelength of 193 nm (ArF), a wavelength of 248 nm (KrF), a wavelength of 365 nm (I-line), or a wavelength of 436 nm (G-line).

The substrate stage **20** may be provided below the light source **10** and the projection optical system **40**. The substrate stage **20** may receive a substrate **W**. The substrate stage **20** may move the substrate **W** in a horizontal direction. The substrate **W** may include a silicon wafer, but the present inventive concept is not limited thereto.

The illumination optical system **30** may be provided between the light source **10** and the substrate stage **20**. The illumination optical system **30** may be provided between the light source **10** and the reticle **50**. For example, the illumination optical system **30** may include an aperture and a lens. The illumination optical system **30** may be configured to carry out an enlargement projection of the light **110** and/or a reduction projection of the light **110**. Alternatively, the illumination optical system **30** may include a mirror, but the present inventive concept is not limited thereto. The illumination optical system **30** may change a path of the light **110** and collimate the light **110**.

The projection optical system **40** may be provided between the reticle **50** and the substrate stage **20**. The projection optical system **40** may focus light **110** on the substrate **W**, whereby the focused light **110** penetrates the reticle **50**. The projection optical system **40** may include an objective lens (e.g., convex lens).

Reticle **50** may be provided between the illumination optical system **30** and the projection optical system **40**. The reticle **50** may be provided in the reticle stage **60**. For example, the reticle **50** may be a transmissive photomask. The reticle **50** may project the light **110** on the substrate **W**.

The reticle stage **60** may receive the reticle **50**. The reticle stage **60** may allow the reticle **50** to move in response to a control signal of a controller (See **102** of FIG. **3**).

FIGS. **2** and **3** illustrate the reticle stage **60** as depicted in FIG. **1**, according to one embodiment.

Referring to FIGS. **2** and **3**, the reticle stage **60** may include first lower plates **62**, first upper plates **64**, a first actuator **66**, a first cable slab **72**, a second cable slab **74**, a first cable cover **82**, and a second cable cover **84**.

The first lower plates **62** may be correspondingly provided below the first upper plates **64**. Although not shown in FIGS. **2** and **3**, the first lower plates **62** may be provided on opposite sides of the first actuator **66**. The first lower plate **62** may guide a lower portion of the second cable cover **84**, the first cable cover **82**, the second cable slab **74**, and the first cable slab **72**. The first lower plate **62** may be a lower guide plate. The first lower plate **62** may move along the first actuator **66**. Alternatively, the first lower plate **62** may stop independently of the first actuator **66**, but the present inventive concept is not limited thereto.

The first upper plates **64** may be provided on opposite sides of the first actuator **66**. The first upper plates **64** may be correspondingly provided on the first lower plates **62**. The first upper plate **64** may be spaced apart from the first lower plate **62**. The first upper plate **64** may guide an upper portion of the second cable cover **84**, the first cable cover **82**, the second cable slab **74**, and the first cable slab **72**. The first upper plate **64** may be an upper guide plate.

The first actuator **66** may be connected to the first lower plate **62**. The first actuator **66** may move along a guide rail (not shown). The guide rail may be parallel to the first lower plate **62** and the first upper plate **64**. For example, the first actuator **66** may include a first long stroke actuator **65** and a first short stroke actuator **67**.

The first long stroke actuator **65** may be provided below the first short stroke actuator **67**. As shown in the plan view of FIG. **2**, the first long stroke actuator **65** may be wider than the first short stroke actuator **67**. The first long stroke actuator **65** may drive the first short stroke actuator **67** to move in a horizontal direction. The first short stroke actuator **67** may be provided on a center of the first long stroke actuator **65**. The first short stroke actuator **67** may have a reticle chuck **69** in a center thereof.

The reticle **50** may be provided in the reticle chuck **69**. The first short stroke actuator **67** may drive the reticle **50** to move in a horizontal direction (e.g., X and Y directions).

The first cable slab **72**, as shown in FIG. **3**, may be provided between the first lower plate **62** and the first upper plate **64**. A controller **102** may be connected through the first cable slab **72** to one side of the first long stroke actuator **65** of the first actuator **66**. The first cable slab **72** may be curved or bent.

Although not shown, each of the first and second cable slabs **72** and **74** may include conductive lines and a cover that surrounds outer peripheries of the conductive lines.

The second cable slab **74** may be provided between the first lower plate **62** and the first upper plate **64**. A controller **102** may be connected through the second cable slab **74** to another side of the first long stroke actuator **65** of the first actuator **66**. The second cable slab **74** may be bent in a direction toward the first cable slab **72**.

The first cable cover **82** may be provided on and below the first cable slab **72**. The first cable cover **82** may protect the first cable slab **72**. When the first cable slab **72** bends, the first cable cover **82** may prevent or minimize abrasion of the first cable slab **72** against the first lower plate **62**. The first cable cover **82** may include Teflon. The first cable cover **82** may not be attached to, but instead be separated from the first cable slab **72**. For example, the first cable cover **82** may have a width of about 280 mm and a length of about 680 mm.

Each of the first and second cable covers **82**, **84** may include a cable jacket and a cable carrier, but the present inventive concept is not limited thereto. In addition, each of the first and second cable covers **82**, **84** may be provided on upper and lower sides of a tube slab (not shown), thereby protecting the tube slab.

The second cable cover **84** may be provided on and below the second cable slab **74**. The second cable cover **84** may protect the second cable slab **74**. When the second cable slab **74** bends, the second cable cover **84** may prevent or minimize abrasion of the second cable slab **74** against the first lower plate **62**. The second cable cover **84** may be the same as the first cable cover **82**. For example, the second cable cover **84** may include Teflon.

Each of the first and second cable slabs **72** and **74** in the first and second cable covers **82** and **84** may have a stack structure or a parallel structure. Likewise, the tube slab may have a stack or parallel structure in the first cable cover **82** and the second cable cover **84**.

Although not shown in FIGS. **2** and **3**, in one embodiment, the first cable cover **82** and the second cable cover **84** may not be coupled to fixing clamps of the first cable slab **72** and the second cable slab.

5

FIGS. 4 and 5 illustrate development and plan views of the first cable cover 82 depicted in FIGS. 2 and 3.

Referring to FIGS. 4 and 5, the first cable cover 82 may include a lower cover 81 and an upper cover 83.

The lower cover 81 may be provided below the first cable slab 72. For example, the lower cover 81 may have a thickness of about 0.75 mm. The lower cover 81 may have folding slits 85. The folding slits 85 may expose a portion of the first cable slab 72. Each of the folding slits 85 may have a rectangular shape when viewed in plan. Each of the folding slits 85 may have a vertical length of about 380 mm and a horizontal length of about 10 mm. The folding slits 85 may have therebetween a distance of about 50 mm to about 100 mm. In addition, the lower cover 81 may have a lower adhesive layer 88 on an edge of a top surface thereof. The lower adhesive layer 88 may include epoxy, but the present inventive concept is not limited thereto.

The upper cover 83 may be provided on the first cable slab 72. For example, the upper cover 83 may have a thickness of about 0.75 mm. The upper cover 83 may have folding slits 85. The upper cover 83 may have an upper adhesive layer 89 on an edge of a bottom surface thereof. The upper adhesive layer 89 may include epoxy.

The lower cover 81 and the upper cover 83 may be replaced in case of emergency or worn out. A knife (not shown) or a pair of scissors (not shown) may be used to separate worn-out lower and upper covers 81, 83 may be separated from the first cable slab 72 or the second cable slab 74. A pair of scissors may be provided on edges of the lower and upper covers 81 and 83 that are on an outer periphery of either the first cable slab 72 or the second cable slab 74. After removing the worn-out lower and upper covers 81, 83, with new lower and upper covers 81, 83 may be provided on the first cable slab 72 or the second cable slab 74. A hand press tool may be utilized to compress and attach the lower cover 81 and the upper cover 83. The hand press tool may heat and compress the lower cover 81 and the upper cover 83.

Each of the lower and upper covers 81, 83 may have handlers 87. The handlers 87 may be provided on edges of each of the lower and upper covers 81 and 83. Each of the handlers 87 may have a tetragonal shape when viewed in plan. Each of the handlers 87 may have a vertical length of about 30 mm and a horizontal length of about 10 mm. The handlers 87 may be usefully employed to align the lower cover 81 and the upper cover 83. In addition, the handlers 87 may be helpfully used to couple the lower cover 81 and the upper cover 83.

The handlers 87 may be removed when the lower and upper covers 81 and 83 are coupled, but the present inventive concept are not limited thereto. As shown in FIG. 5, the handlers 87, the lower adhesive layer 88, and the upper adhesive layer 89 may be omitted on an edge of the first cable slab 72 between the lower cover 81 and the upper cover 83.

Although not shown in FIGS. 4 and 5, the second cable cover 84 may include a lower cover 81 and an upper cover 83, but the present inventive concept is not limited thereto.

FIG. 6 is a cross-sectional view, the reticle stage 60 depicted in FIG. 1, according to an embodiment.

Referring to FIG. 6, the reticle stage 60 may further include a roller vehicle 90. The roller vehicle 90 may be provided on the first and second cable slabs 72 and 74 that are curved. For example, the roller vehicle 90 may include a first roller 92, a second roller 94, and a connection bar 96.

The first roller 92 may be provided on the first cable slab 72 and the first cable cover 82. For example, the first roller

6

92 may include a spring roll. Alternatively, the first roller 92 may include a bearing roller, but the present inventive concept is not limited thereto. The first roller 92 may be provided on a curved portion of the first cable slab 72. For example, the first roller 92 may include a spring roll.

The second roller 94 may be provided on the second cable slab 74 and the second cable cover 84. The second roller 94 may be provided on a curved portion of the second cable slab 74. The second roller 94 may be the same as the first roller 92. For example, the second roller 94 may include a spring roll.

The connection bar 96 may connect the first roller 92 to the second roller 94. The connection bar 96 may define a distance between the first roller 92 and the second roller 94. The connection bar 96 may move together with the first and second rollers 92 and 94.

Therefore, the roller vehicle 90 may maintain a constant distance between the first cable slab 72 and the second cable slab 74.

FIG. 7 is a cross-sectional view of the roller vehicle 90 depicted in FIG. 6, according to an embodiment.

Referring to FIG. 7, the roller vehicle 90 may be provided on and below each of the first cable slab 72 and the second cable slab 74. The first roller 92 of the roller vehicle 90 may be provided on and below the first cable slab 72. For example, the first roller 92 may include a first upper roller 91 and a first lower roller 93.

The first upper roller 91 may be provided on the first cable slabs 72 and the first cable cover 82. The first upper roller 91 may be provided in curved or bent portions of the first cable slab 72 and the first cable cover 82. The first upper roller 91 may have a diameter greater than that of the first lower roller 93.

The first lower roller 93 may be provided below the first cable slab 72 and the first cable cover 82. The first lower roller 93 may support curved portions of the first cable slab 72 and the first cable cover 82. When the first cable slab 72 and the first cable cover 82 become bent, the first lower roller 93 may reduce loads of the first cable slab 72 and the first cable cover 82 against the first lower plate 62. In addition, the first lower roller 93 may reduce a contact area between the first cable cover 82 and the first lower plate 62, thereby decreasing abrasion of the first cable cover 82. A rotational direction of the first lower roller 93 may be opposite to that of the first upper roller 91.

The second roller 94 of the roller vehicle 90 may be provided on and below each of the second cable slabs 74 and the second cover 84. For example, the second roller 94 may include a second upper roller 95 and a second lower roller 97.

The second upper roller 95 may be provided on the second cable slab 74 and the second cable cover 84. The second upper roller 95 may be provided in curved portions of the second cable slab 74 and the second cable cover 84. The second upper roller 95 may have a diameter greater than that of the second lower roller 97.

The second lower roller 97 may be provided below the second cable slab 74 and the second cable cover 84. The second lower roller 97 may support curved portions of the second cable slab 74 and the second cable cover 84. When the second cable slab 74 and the second cable cover 84 become bent, the second lower roller 97 may reduce loads of the second cable slab 74 and the second cable cover 84 against the first lower plate 62. In addition, the second lower roller 97 may reduce a contact area between the second cable cover 84 and the first lower plate 62, thereby decreasing abrasion of the second cable cover 84. A rotational direction

of the second lower roller **97** may be opposite to that of the second upper roller **95** and identical to that of the first lower roller **93**.

The first upper roller **91**, the first lower roller **93**, the second upper roller **95**, and the second lower roller **97** may be connected through the connection bar **96** to each other in a tetragonal shape. For example, the connection bar **96** may include a first level connection bar **98**, a first horizontal connection bar **99**, a second level connection bar **111**, and a second horizontal connection bar **113**.

The first level connection bar **98** may connect the first upper roller **91** to the first lower roller **93**. The first level connection bar **98** may define a distance between the first upper roller **91** to the first lower roller **93**. The first level connection bar **98** may be inclined to the first lower plate **62**. Alternatively, the first level connection bar **98** may be perpendicular to the first lower plate **62**, but the present inventive concept is not limited thereto.

The first horizontal connection bar **99** may connect the first upper roller **91** to the second upper roller **95**. The first horizontal connection bar **99** may define a distance between the first upper roller **91** to the second upper roller **95**. The first horizontal connection bar **99** may maintain a constant distance between curved portions of the first and second cable slabs **72** and **74**.

The second level connection bar **111** may connect the second upper roller **95** to the second lower roller **97**. The second level connection bar **111** may define a distance between the second upper roller **95** to the second lower roller **97**. The second level connection bar **111** may be inclined relative to the first lower plate **62**. Alternatively, the second level connection bar **111** may be perpendicular to the first lower plate **62**, but the present inventive concept is not limited thereto.

The second horizontal connection bar **113** may connect the second lower roller **97** to the first lower roller **93**. The second horizontal connection bar **113** may define a distance between the first lower roller **93** to the second lower roller **97**. The second horizontal connection bar **113** may adjust a contact area between the first cable cover **82** and the first lower plate **62** and may also adjust a contact area between the second cable cover **84** and the first lower plate **62**.

According to the present inventive concept, the semiconductor fabrication apparatus **100** may be configured such that the roller vehicle **90** is used to minimize abrasion of the first cable cover **82** and the second cable cover **84**.

FIG. **8** is a view illustrating the reticle stage **60** depicted in FIG. **1**, according to an embodiment.

Referring to FIG. **8**, the reticle stage **60** may further include first outer plates **61** and first inner plates **63**. The first actuator **66** may be configured identically to that of FIG. **2**.

The first outer plates **61** may be provided outside opposite sidewalls of the first actuator **66**. The first outer plates **61** may be provided on outer peripheries of the first cable cover **82** and the second cable cover **84**.

The first inner plates **63** may be provided between the first outer plates **61** and the opposite sidewalls of the first actuator **66**. The first inner plates **63** may be parallel to the first outer plates **61**.

The first cable cover **82** and the second cable cover **84** may be provided between the first outer plates **61** and the first inner plates **63**. The first cable cover **82** and the second cable cover **84** may be folded in a direction parallel to the first actuator **66**. For example, the first cable cover **82** and the second cable cover **84** may be folded in a horizontal direction. Although not shown in FIG. **8**, the first cable slab

(see **72** of FIG. **3**) and the second cable slab (see **74** of FIG. **3**) may be folded in a horizontal direction.

Referring again to FIG. **1**, the substrate stage **20** may be configured identically to the reticle stage **60** of FIGS. **2**, **3**, and **6** to **8**, but the present inventive concept is not limited thereto.

FIG. **9** illustrates the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concept.

Referring to FIG. **9**, a semiconductor fabrication apparatus **100** of the present inventive concept may be an extreme ultraviolet (EUV) exposure device. For example, the semiconductor fabrication apparatus **100**, according to the present embodiment, may further include a chamber **12**.

The chamber **12** may envelope the light source **10**, the substrate stage **20**, the illumination optical system **30**, the projection optical system **40**, the reticle **50**, and the reticle stage **60**. The chamber **12** may store or hermetically seal plasma (not shown) of the light source **10**.

The light source **10** may be provided on one side of a lower portion of the chamber **12**. The light **10** may generate either plasma or light **110** of plasma beam. For example, the light source **10** may produce the light **110** by providing pump light to liquid metal droplets of tin (Sn), xenon (Xe), titanium (Ti), or lithium (Li). The light **110** may include an extremely ultraviolet (EUV) beam having a wavelength of about 13.5 nm.

The substrate stage **20** may be provided on a center of the lower portion of the chamber **12**. The substrate stage **20** may be provided below the illumination optical system **30** and the projection optical system **40**. The substrate stage **20** may receive and move the substrate W.

FIG. **10** illustrates the substrate stage **20** of the semiconductor fabrication apparatus as depicted in FIG. **9**.

Referring to FIG. **10**, the substrate stage **20** may include a second lower plate **22**, a second upper plate **24**, a second actuator **26**, a chucking plate **28**, a third cable slab **76**, and a third cable cover **86**.

The second lower plate **22** may be provided below the second upper plate **24**, the second actuator **26**, and the chucking plate **28**. The second lower plate **22** may be connected to the controller **102**. Alternatively, the controller **102** may be separated from the second lower plate **22**, but the present inventive concept is not limited thereto.

The second upper plate **24** may be provided on the second lower plate **22**. The second upper plate **24** may be parallel to the second lower plate **22**. The second upper plate **24** may be utilized as a cover plate.

The second actuator **26** may be coupled to the second upper plate **24**. The second actuator **26** may move along a second guide rail (not shown). The second guide rail may be parallel to the second upper plate **24**. The second actuator **26** may support the chucking plate **28**. The second actuator **26** may drive the chucking plate **28** to move upwards, downwards, leftwards, and rightwards.

Although not shown in FIG. **10**, the second actuator **26** may include a second long stroke actuator and a second short stroke actuator. The second long stroke actuator may be provided below the second short stroke actuator. The second long stroke actuator may be wider than the second short stroke actuator. The second long stroke actuator may drive the second short stroke actuator to move in a horizontal direction. The second short stroke actuator may be provided on a center of the second long stroke actuator. The second short stroke actuator may have a substrate chuck in a center thereof. The substrate W may be provided on the substrate

chunk. The first short stroke actuator may drive the substrate W to move in a horizontal direction (e.g., X and Y directions).

The chucking plate 28 may be provided on the second upper plate 24 and the second actuator 26. The chucking plate 28 may receive the substrate W. The chucking plate 28 may fix the substrate W, and the second actuator 26 may drive the chucking plate 28 to move the substrate W.

The third cable slab 76 may connect the second actuator 26 to the controller 102. The third cable slab 76 may be provided between the second lower plate 22 and the second upper plate 24. The third cable slab 76 may be curved or bent. A curved portion of the third cable slab 76 may move together with the second actuator 26. Although not shown in FIG. 10, the third cable slab 76 may be provided in plural in a horizontal direction (e.g., X and Y directions).

The third cable cover 86 may surround an outer circumferential surface of the third cable slab 76. The third cable cover 86 may protect the third cable slab 76. The third cable cover 86 may be resistant to plasma. Although not shown, the third cable cover 86 may include a lower cover and an upper cover. The lower cover may be provided below the third cable slab 76. The upper cover may be provided on the third cable slab 76. The lower and upper covers may be coupled and/or attached to each other on an outer periphery of the third cable slab 76. Each of the lower and upper covers may have folding slits. The third cable cover 86 may be provided in plural in a horizontal direction (e.g., X and Y directions).

Referring again to FIG. 9, the illumination optical system 30 may be provided between the light source 10 and the reticle 50. The illumination optical system 30 may reflect the light 110 on the reticle 50. The illumination optical system 30 may include a plurality of flat mirrors.

The projection optical system 40 may be provided between the reticle 50 and the substrate stage 20. The projection optical system 40 may be configured such that the light 110 is reflected to project an image of the reticle 50 on the substrate W. The projection optical system 40 may include one or more of concave mirrors, convex mirrors, and flat mirrors, but the present inventive concept is not limited thereto.

The reticle 50 may be provided in a central upper portion of the chamber 12. Although not shown, the reticle 50 may be provided on the reticle stage 60, while being received in a reticle stocker in the chamber 12. The reticle 50 may reflect the light 110 on the projection optical system 40.

The reticle stage 60 may be provided in a central upper portion of the chamber 12. The reticle stage 60 may fix the reticle 50. The reticle stage 60 may be configured identically to the substrate stage 20. In addition, the reticle stage 60 may be configured identically to that of FIGS. 2, 3, and 6 to 8, but the present inventive concept is not limited thereto.

According to the present inventive concept, the semiconductor fabrication apparatus 100 may be configured such that the first cable cover 82, the second cable cover 84, and the third cable cover 86 are used to prevent or minimize abrasion of the first cable slab 72, the second cable slab 74, and the third cable slab 76.

As discussed above, a semiconductor fabrication apparatus according to some embodiments of the present inventive concept may be configured such that a cable cover surrounding a cable slab is used to prevent or minimize abrasion of the cable slab.

The above descriptions are specific examples for practicing the present inventive concept. The present inventive concept will include not only the embodiments described

above but also embodiments that can be easily or simply changed in design. In addition, the present inventive concept will also include technique that can be easily modified and implemented using the embodiments described above.

What is claimed is:

1. A substrate fabrication apparatus, comprising:

a light source configured to emit light;

a substrate stage configured to receive a substrate exposed to the light;

a reticle disposed between the substrate stage and the light source; and

a reticle stage configured to receive the reticle, wherein the reticle stage includes:

lower plate;

an upper plate disposed on the lower plate;

an actuator connected to the lower plate and configured to move in a direction parallel to the upper plate;

a first cable slab disposed between the upper plate and the lower plate, the first cable slab connected to a first side of the actuator; and

a first cable cover surrounding an outer periphery of the first cable slab, the cable cover configured to contact the lower plate when the first cable slab is bent.

2. The substrate fabrication apparatus of claim 1, wherein the first cable cover includes:

a lower cover disposed below the first cable slab; and

an upper cover disposed on the first cable slab and on an edge of the lower cover on an outer periphery of the first cable slab.

3. The substrate fabrication apparatus of claim 2, wherein each of the upper cover and the lower cover includes a plurality of folding slits that expose portions of the first cable slab.

4. The substrate fabrication apparatus of claim 3, wherein each of the plurality of folding slits has a rectangular shape.

5. The substrate fabrication apparatus of claim 1, wherein the reticle stage further includes:

a second cable slab disposed between the upper plate and the lower plate, the second cable slab connected to a second side of the actuator; and

a second cable cover surrounding an outer periphery of the second cable slab, the second cable cover configured to contact the lower plate when the second cable slab is bent.

6. The substrate fabrication apparatus of claim 5, wherein the reticle stage further includes a roller vehicle disposed on the first cable slab and the second cable slab, the roller vehicle

configured to move together with the actuator.

7. The substrate fabrication apparatus of claim 6, wherein the roller vehicle includes:

a first roller disposed on the first cable cover;

a second roller disposed on the second cable cover; and a connection bar configured to connect the first roller to the second roller.

8. The substrate fabrication apparatus of claim 7, wherein the first roller includes:

a first upper roller disposed on the first cable cover; and

a first lower roller disposed below the first cable cover.

9. The substrate fabrication apparatus of claim 8, wherein the second roller includes:

a second upper roller disposed on the second cable cover; and

a second lower roller disposed below the second cable cover.

10. The substrate fabrication apparatus of claim 9, wherein the connection bar includes:

11

an upper connection bar disposed between the first upper roller and the second upper roller;
 a first connection bar disposed between the first upper roller and the first lower roller;
 a lower connection bar disposed between the first lower roller and the second lower roller; and
 a second connection bar disposed between the second upper roller and the second lower roller.

11. A semiconductor fabrication apparatus, comprising:
 a chamber;
 a light source in the chamber configured to emit light;
 a substrate stage configured to receive a substrate exposed to the light;
 a reticle disposed between the substrate stage and the light source; and
 a reticle stage configured to receive the reticle, wherein the substrate stage includes:
 a lower plate;
 an upper plate disposed on the lower plate;
 an actuator connected to the lower plate moving in a direction parallel to the upper plate;
 a cable slab between the upper plate and the lower plate, the cable slab being connected to a first side of the actuator; and
 a cable cover surrounding an outer periphery of the cable slab, the cable cover configured to contact the lower plate when the cable slab is bent.

12. The semiconductor fabrication apparatus of claim **11**, wherein the cable cover includes Teflon.

13. The apparatus of claim **11**, wherein the cable cover includes:
 a lower cover disposed below the cable slab; and
 an upper cover disposed on the cable slab and an edge of the lower cover on an outer periphery of the cable slab.

14. The substrate fabrication apparatus of claim **13**, wherein each of the upper cover and the lower cover has a plurality of folding slits exposing portions of the cable slab.

15. The substrate fabrication apparatus of claim **14**, wherein each of the folding slits has a rectangular shape.

16. A semiconductor fabrication apparatus, comprising:
 a light source configured to emit light;
 a substrate stage configured to receive a substrate exposed to the light emitted from the light source;
 a reticle disposed between the substrate stage and the light source; and
 a reticle stage configured to receive the reticle, wherein the reticle stage includes:

12

a first lower plate;
 a first upper plate disposed on the first lower plate;
 a first actuator connected to the first lower plate and configured to move in a direction parallel to the first upper plate;
 a first cable slab disposed between the first upper plate and the first lower plate, the first cable slab connected to a first side of the first actuator;
 a first cable cover surrounding an outer periphery of the first cable slab configured to contact the first lower plate when the first cable slab is bent;
 a second cable slab disposed between the first upper plate and the first lower plate connected to a second side of the first actuator;
 a second cable cover surrounding an outer periphery of the second cable slab configured to contact the first lower plate when the second cable slab is bent; and
 a roller vehicle disposed on the first cable slab and the second cable slab, the roller vehicle configured to move together with the first actuator.

17. The semiconductor fabrication apparatus of claim **16**, wherein the roller vehicle includes:
 a first roller disposed on the first cable cover;
 a second roller disposed on the second cable cover; and
 a connection bar configured to connect the first roller to the second roller.

18. The semiconductor fabrication apparatus of claim **17**, wherein the first roller includes:
 a first upper roller disposed on the first cable cover; and
 a first lower roller disposed below the first cable cover.

19. The semiconductor fabrication apparatus of claim **18**, wherein the second roller includes:
 a second upper roller disposed on the second cable cover; and
 a second lower roller disposed below the second cable cover.

20. The semiconductor fabrication apparatus of claim **19**, wherein the connection bar includes:
 an upper connection bar disposed between the first upper roller and the second upper roller;
 a first connection bar disposed between the first upper roller and the first lower roller;
 a lower connection bar disposed between the first lower roller and the second lower roller; and
 a second connection bar disposed between the second upper roller and the second lower roller.

* * * * *